



Design and implementation of low complexity circularly symmetric 2D FIR filter architectures

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Abstract

This paper presents a low complexity two dimensional (2D) circular symmetric Finite Impulse Response (FIR) filter design and implementation of architecture. The optimized circular symmetric 2D FIR filter is designed using a modified Park–McClellan transformation method and filter coefficients are quantized using a canonical signed digit (CSD) binary number format. The CSD encoded coefficients are optimized to reduce the number of adder/subtractors using common subexpression elimination (CSE) algorithms. Based on the modified filter coefficients, the two structures fully direct form and hybrid-II form 2D FIR filter architectures are implemented using CSD and CSD with Horizontal CSE and Vertical CSE techniques. The proposed architectures compared with the conventional symmetry 2D filters and state-of-the-art architectures in terms of area, power, and speed.

Keywords 2D FIR filter · McClellan transformation · Canonical signed digit (CSD) · Multiple constant multiplication (MCM) · Common sub-expression elimination (CSE) · Low power

1 Introduction

The two-dimensional (2D) linear phase filters are frequently used in image and video processing applications. The efficient and low complexity Finite Impulse Response (FIR) filters often need for image restoration, image enhancement, and denoising applications, etc. The design and implementation of low power and low complexity with high-speed digital filter architecture is a challenging task. The optimization is required for 2D FIR filters to meet the coveted filter specifications. The pruning of hardware complexity with low power consumption is desired for the Very Large Scale Integration (VLSI) implementation of any filter architecture.

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Many optimization algorithms are proposed for designing the 2D FIR filters, such as Genetic Algorithm, Particle Swarm and Differential Evolution, ABC algorithm, etc. (Chandra and Chattopadhyay 2016). The above methods suffer from the large execution time and premature convergence. The optimization is carried out to meet the filter frequency domain specifications, passband ripple, and stopband ripple. The McClellan transformation method is the best technique for the design of 2D FIR filters for an efficient hardware structure implementation. McClellan transformation based method results in a low complexity 2D FIR architecture using a transformation approach. The circularly symmetric, fan type and elliptic filters are designed by kth-McClellan transformation technique with remarkable results. The transformation approach is most valuable for linear phase filters and they should be in the symmetry of quadrant. This algorithm is fast and it gives a moderate degree and good implementation. Here, the one dimensional (1-D) FIR filter is designed and mapped to 2D FIR Filter using McClellan transform (Das et al. 2019). The kth-McClellan transform is added by a new term which is used to get circular shape at all frequencies is called as T1 transformation. The T1 transformation is modified into to approximate the constants to avoid the squarish shape at the center of the low pass filter is T2 transformation. The T2 is changed to P2 transformation to achieve circularity for wideband frequencies in 2D FIR filters (Bindima et al. 2016).

Recently, Bindima and Elias (2016, 2017, 2019) proposed low complexity circular and fan type 2D FIR filter design methods using tunable Farrow structure. This work focused on the design of higher-order filters using P2 transformation with Parks McClellan transformation. The design required a huge number of multipliers for the implementation of the hardware architecture. In Baboji and Sridevi (2019), Frequency Response Masking (FRM) approach is proposed for the multiplierless design of 1-D FIR filters with low complexity in terms of hardware. The Signed Sum of Powers of Two (SOPOT) form is used to approximate the filter coefficients with less run time. The Modified Artificial Bee-Colony algorithm (MABC) is proposed in Dwivedi et al. (2016), to optimize the coefficients of circular symmetric 2D FIR filters. This design is validated and synthesized to reduce the power consumption, passband ripple, and stopband ripple. Many authors have proposed 2D FIR filter design concepts and optimization techniques to reduce the complexity and power (Odugu et al. 2019). Here, the optimization is carried by many transformations in the design level only and no exploration in architecture level for 2D FIR filters.

The designed filter implementation in hardware with necessary optimization schemes is a challenging task. Many efficient multipliers are available for low complexity implementation of FIR filters. But, the full flexibility is not required for the multiplication process in filters and multiplier is a power-hungry circuit in the filter architecture. To multiply the inputs and delay version of inputs with fixed filter coefficient constants, Multiple Constant Multiplication (MCM) technique and multiplierless filter operations can be used.

Many systolic 2D filter architectures are proposed with less number of computations by symmetry property. Mohanty et al. (2013) proposed, the memory-efficient 2D FIR filters inseparable and non-separable structures with medium critical path and medium computations. But in this work, multipliers are complex blocks and occupies more area and consumes large power.

The symmetry property in the frequency response of the filter reduces the number of multipliers or computations. In the papers (Chen et al. 2010, 2013), area and power-efficient 2D filter structures with various symmetry properties are suggested with complex multipliers. Due to the symmetry number of multipliers are reduced but, these multipliers are power-consuming blocks in the filter. Aksoy et al. (2014) have been introduced efficient multiplierless designs with the minimum number of adders/subtractors for only one dimen-

sional FIR filters using exact and approximated models. Recently, a Distributed Arithmetic (DA) block-based 2D FIR filter in a systolic structure style is proposed using pipelining and parallel processing to reduce the complexity of hardware (Kumar et al. 2018, 2019).

Many researchers are focused on 2D filter architectures with no exploration of design issues. Whereas designers are only concentrated on the design concepts and optimization in filter specifications and no exploration for the hardware structures.

In this paper, at the design level, the filter coefficients are optimized and hardware architectures are implemented using appropriate optimizing algorithms. The design of 2D FIR filter using T2 and P2 transformations in kth-McClellan transform gives circularity in the contour of low pass filter. The efficient approach for the implementation of designed coefficients of 2D FIR filter is canonical signed digit (CSD) encoding method under multiplierless filter implementation. The CSD format consists of less number of non zero digits and causes to reduce the number of adders/subtractors. The CSD coefficients of the filter will be implemented using add and shift operations only.

In this paper, the 2D FIR filter is designed with circular symmetry using kth-Park McClellan transform using T2 and P2 transformations for desired specifications and different VLSI architectures are suggested with low complexity. The optimized 1-D FIR filter is designed and transformed to 2D filter using a transformation approach. Next, the designed filter coefficients are optimized using CSD, CSD with horizontal common subexpression elimination (HCSE) and CSD with vertical common subexpression elimination (VCSE) techniques. Based on the optimized coefficients, the hardware architectures are implemented in the form of fully direct type and hybrid-II type filters.

The paper is organized as follows: Sect. 2 presents the design of circular symmetry 2D FIR filter using T2 and P2 transformations with CSD space. The word length and number of Signed Power (SPT) terms are restricted without any deviation from given specifications. Section 3 provides the 2D FIR filter architectures in fully direct form and hybrid-II form architectures for continuous coefficients, CSD coefficients and optimized CSD coefficients using HCSE and VCSE. Experimental results and comparison of hardware complexity of different architectures styles are presented in Sect. 4. The work conclusion is summarized in Sect. 5.

2 Design of 2D circularly symmetric FIR filter

The McClellan transform is used to transform the zero phase 1-D FIR filter into zero phase 2D FIR filter. This transformation mostly applied for odd length one-dimensional filters. The length of the 1-D filter with zero phase is $2N + 1$ and it should be symmetric. The frequency response of the 1-DFIR filter is given in Eq. 1.

$$H(e^{j\omega}) = h(0) + \sum_{n=1}^N h(n)[e^{-j\omega n} + e^{j\omega n}] \quad (1)$$

$$= h(0) + \sum_{n=1}^N 2h(n)\cos\omega n \quad (2)$$

where $h(n)$ is an impulse response. Then 1-D zero-phase filter frequency response is converted into 2D filter response with zero phase and is denoted by Eq. 2. The McClellan transforma-

tion generally transforms the 1-D to 2D filter using the substitution of some variables. The substitution term is given by Eqs. 3 and 4.

$$H(e^{j\omega_1}, e^{j\omega_2}) = \sum_{m=0}^{M1} \sum_{n=0}^{M2} a(m, n) \cos m\omega_1 \cos n\omega_2 \quad (3)$$

$$\cos \omega = \sum_{p=0}^P \sum_{q=0}^Q t(p, q) \cos p\omega_1 \cos q\omega_2 \quad (4)$$

Finally, the relation of 2D impulse response $h(m, n)$ and 1-D filter impulse response $h(n)$ is can be written as Eq. 5,

$$H(e^{j\omega}) = \sum_{n=0}^N b(n) [\cos \omega]^n \quad (5)$$

According to the Chebyshev polynomial recurrence formula, the overall response of 2D FIR filter is written in the form of Eq. 2 with filter sizes $(2NP + 1) \times (2NQ + 1)$ expressed in Eq. 6.

$$H(e^{j\omega_1}, e^{j\omega_2}) = \sum_{n=0}^N b(n) \left[\sum_{p=0}^P \sum_{q=0}^Q t(p, q) \cos p\omega_1 \cos q\omega_2 \right]^n \quad (6)$$

Here, the McClellan transform considered for the case of $P = 1$ and $Q = 1$ only. The shape of the 2D filter contours depends on the parameter values of $t(p, q)$, where $0 < p < P$, and $0 < q < Q$. The frequency response of the 2D FIR filter is controlled by the impulse response of 1-D filter $h(n)$. First, a 1-D prototype FIR filter is designed for the desired specifications and transformed into a 2D FIR filter using McClellan. In this paper, a circular symmetric filter is considered with parameter values of $P = 1$ and $Q = 1$, then $-t(0, 0) = t(1, 0) = t(0, 1) = t(1, 1) = 1/2$.

By appropriately selecting the $t(p, q)$ values different shapes of filters can be designed. Such as circular, elliptical, diamond and fan shapes, etc. This original McClellan transformation gives circular contour for small radii values and it nearly becomes square-shaped at higher radii frequencies of ω . For low pass filter, it gives square shape at the center of the contour as shown in the Fig. 1. The contour has to be approximated using appropriate transformation parameters $t(p, q)$ to achieve exact circular shape at the center of the low pass filter. The contours are produced by the transformation by approximating the constants in the passband and stopband of the low pass filter (Liu and Tai 2010; Shyu et al. 2008). For approximation consider kth-McClellan transform as given in Eqs. 7 and 8.

$$\cos \omega = F(\omega_1, \omega_2) \quad (7)$$

$$F(\omega_1, \omega_2) = \sum_{p=0}^P \sum_{q=0}^Q t(p, q) \cos p\omega_1 \cos q\omega_2 \quad (8)$$

The original kth-McClellan transform can be written as Eq. 9 using basic trigonometric series.

$$\cos(\Omega) = f(\omega_1, \omega_2) = 2 \left[\cos\left(\frac{\omega_1}{2}\right) \cos\left(\frac{\omega_2}{2}\right) \right]^2 - 1 \quad (9)$$

$$\cos(\Omega) = g(\omega_1, \omega_2) = 2 \left[\cos\left(\frac{\omega_1}{2}\right) \cos\left(\frac{\omega_2}{2}\right) \right]^{2k} - 1 \quad (10)$$

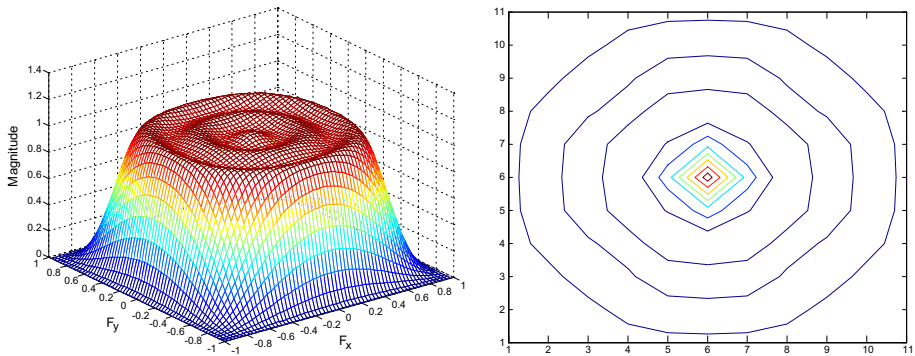


Fig. 1 Frequency response and contour of Low Pass Filter using McClellan transform

where k is integer constant i.e. $k \geq 1$. It is nothing but k th-McClellan transform. If k value is 1, then the above Eq. 10 becomes an original McClellan transform. The frequency response of the low pass filter using k th-McClellan transform is shown in Fig. 1.

The squarish effect at the center of the low pass filter, and for higher frequencies on (ω_1, ω_2) plane can be observed from the Fig. 1. This is overcome by introducing the additional terms for the k th-McClellan transform equation and the modified equations are called T1 transformation as given by Eq. 11 and T2 transformation is given by Eq. 12.

$$\cos(\Omega) = 2 \left[\cos\left(\frac{\omega_1}{2}\right) \cos\left(\frac{\omega_2}{2}\right) \right]^{2k} \times \left[1 - \sin^2\left(\frac{\omega_1}{2}\right) \sin^2\left(\frac{\omega_2}{2}\right) \right] - 1 \quad (11)$$

$$\cos(\Omega) = 2 \left[\cos\left(\frac{\omega_1}{2}\right) \cos\left(\frac{\omega_2}{2}\right) \right]^{2k} \times \left[\cos^2\left(\frac{\omega_1}{2}\right) + \cos^2\left(\frac{\omega_2}{2}\right) \right] - 1 \quad (12)$$

Further, another modified term is introduced in the T2 transformation to improve the circularity at the higher frequency for wideband filters Bindima and Elias (2019). i. e $P(\omega_1, \omega_2) = 0.5[\cos^2(\frac{\omega_1}{2}) + \cos^2(\frac{\omega_2}{2})]$ is called as P2 transformation is given by (13). The frequency response and contour of the circular symmetric 2D FIR filter using P2 transformation in k th-McClellan transform is shown in the Fig. 2.

$$\cos(\Omega) = 2 \left[\cos\left(\frac{\omega_1}{2}\right) \cos\left(\frac{\omega_2}{2}\right) \right]^{2k} \times P(\omega_1, \omega_2) - 1 \quad (13)$$

The smaller order filters are more preferable for image enhancement applications than higher-order filters. Hence to achieve low power and better optimization in hardware complexity, lower-order filters are designed in this work. To obtain a lower order, the following specifications are considered. The proposed 2D FIR filter design is applied for the Low pass filter with specifications of, band edges in Hz = [1500 2000], desired magnitude values in each band is [1 0], the ripple in each band is [0.0559 0.01] and the sampling frequency is 4000 Hz. This specified 2D FIR filter is designed using P2 transformation in the k th-McClellan transform and the order of the filter is obtained as 9×9 with Stop Band (SB) attenuation of 32 dB and Pass Band (PB) ripple 5.8 dB.

The continuous coefficient filter design result in SB attenuation of 32 dB and PB ripple of 5.8 dB. If the filter coefficients are quantized using 18-bit CSD number representation with the precision of 7 SPT terms then it results in an SB attenuation of 30.7 dB and a PB ripple of 6.017 dB. To reduce the number of non-zero bits in CSD, the word length is decreased to 16-bits with SPT terms of 6. It gives the SB attenuation of 29.93 dB and PB ripple of 6.016 dB.

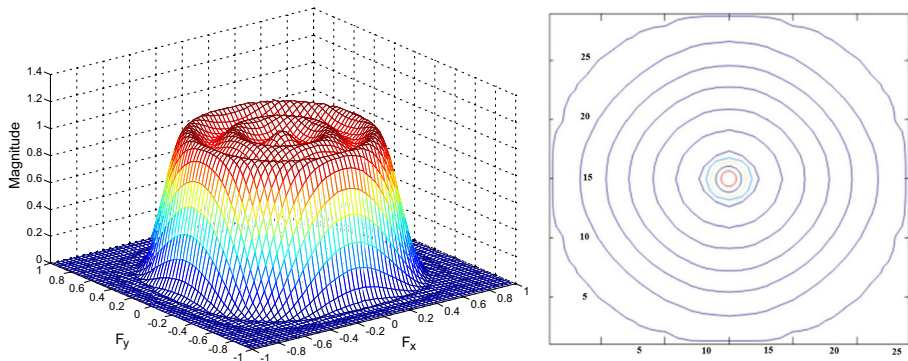


Fig. 2 Frequency response and contour of Low Pass Filter using P2 transformation in kth McClellan transform

Table 1 SB attenuation & PB ripple values for 2D FIR filter for different word lengths and SPT terms

Word length (bits)	No. of SPT terms	Stop band attenuation (dB)	Pass band ripple (dB)
Continuous Coefficient	–	32	5.8
18	7	30.7	6.017
16	6	29.93	6.016
15	6	29.28	6.00
14	5	29.12	6.47
12	5	28.72	6.53

Further, when the coefficients are quantized with less number of SPT terms by decreasing of word length to 12-bits, the SB attenuation is 28.72 dB and PB ripple is 6.53 dB. Table 1 shows the different word lengths and SPT terms with SB and PB results of the designed 2D FIR filter. If the word length is decreased beyond 12-bits with restricted number SPT terms then the performance of the filter is degraded. Hence, a 12-bit CSD coefficient quantization coding is used to reduce the hardware complexity without any diversion of filter parameters. The resultant filter parameters, SB attenuation, and PB ripple are acceptable in range.

The frequency response of the 2D FIR filter with CSD encoding with a word length of 12-bits and 5 SPT terms has not deviated from the magnitude response of the continuous coefficients magnitude response. Hence, the decreasing in word length bits and SPT terms can reduce the number of adders required for the implementation of filter architecture.

3 Implementation of 2D FIR filter architectures

In this section, the 2D FIR filter circular symmetric architecture is implemented in hardware using continuous coefficients with MCM, multiplierless multiplication scheme using CSD. The filter coefficients obtained from the design of the circular symmetric 2D FIR filter using MATLAB for the given specifications are considered for the hardware implementation. The multiple constant multiplications are different concerning the type of architectures. The MCM operation is preferred for the transpose form type filter and Constant Array Vector Multiplication (CAVM) technique is used for the implementation of direct form filters. 2D FIR filters can be implemented in 4 different types of architectures are, (1) fully direct form,

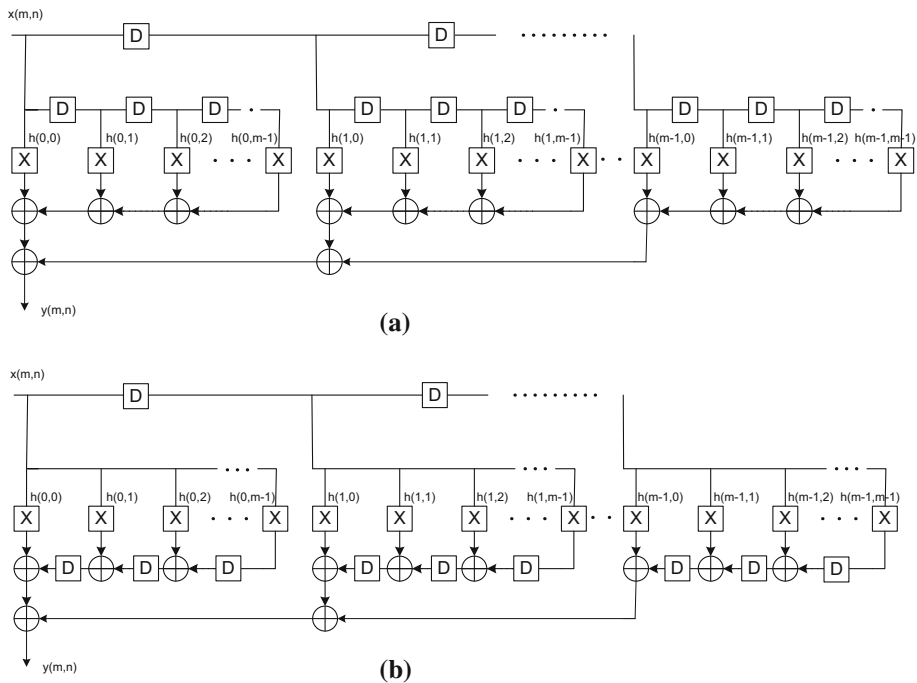


Fig. 3 General Architectures of 2D FIR filters. **a** Fully Direct form **b** Hybrid-II form

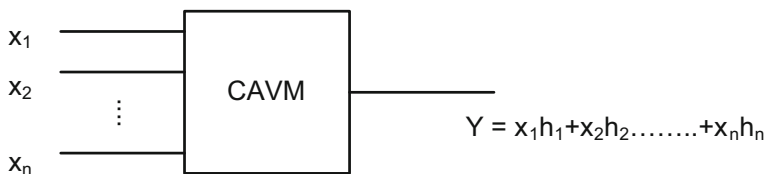


Fig. 4 Basic CAVM technique

(2) fully transpose form, (3) transpose-direct form (hybrid-I) and (4) direct-transpose form (hybrid-II) structures. In this work, fully direct form and hybrid-II structures are implemented with circular symmetry. The symmetrical filters itself reduces some hardware complexity. The general structure of fully direct form and hybrid-II type (direct-transposed type) 2D FIR filters without symmetry is shown in the Fig. 3a, b.

In CAVM, the current input and time-shifted inputs are considered as a vector. The input vector is multiplied with an array of constant coefficients as shown in Fig. 4. The symmetrical 2D FIR filter architectures in fully direct form and hybrid-II form are shown in Fig. 5a, b with CAVM and MCM multiplication techniques respectively.

In the hybrid-II type architecture, the current input sample is multiplied with all coefficients as same as MCM technique. Here, at the input level, the registers will give the delayed inputs for each row filter and internal row filter is implemented in transposed form. The overall filter is in direct form type hence, it is treated as hybrid-II (direct-transpose form) filter.

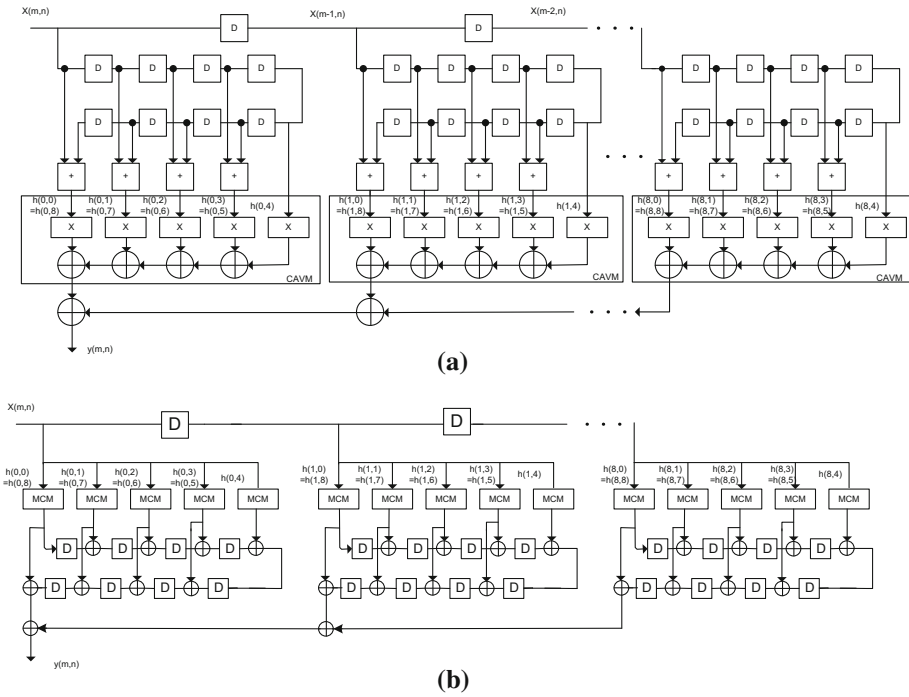


Fig. 5 **a** Fully direct form 2D FIR Filter architecture for continuous coefficients with symmetry using CAVM operation **b** Hybrid-II 2D FIR filter using MCM technique with symmetry of coefficients

3.1 Multiplierless 2D FIR filter architecture implementation using CSD

In multiplierless filter operation, the filter coefficients are denoted by an optimized binary encoding number format for pruning the number of adders/subtractors. The constant-coefficient multiplication scheme is optimized using the CSD representation of coefficients of the filter. The number of adders is reduced to less than the non-zero bits of the coefficient. This representation is used to reduce the area and power consumption of the filter by encoding the coefficients in the fewest number of non-zero elements. This representation is most preferable for high-level synthesis. A coefficient or any number is considered as $C = c_{w-1}, c_{w-2}, \dots, c_1, c_0$. Where each number $c_i (P-1 \geq i \geq 0)$ is in the set of $\{-1, 0, 1\}$ is a ternary number system. CSD representation is a special case and unique number representation, which consist of no two consecutive bits in the number are non-zero. It is possible minimum non-zero number representation is called canonic. The CSD number representation contains about 33% of fewer non zero bits comparatively conventional 2's complement number (Parhi 1999). The iterative algorithm represents the computation of CSD format for the p-bit number.

$$\begin{aligned} \bar{c}_{-1} &= 0 \\ \alpha - 1 &= 0 \\ \bar{c}_p &= \bar{c}_{p-1} \\ \text{for } (i &= 0 \text{ to } p-1) \\ \{\beta_i &= \bar{c}_i \oplus \bar{c}_{i-1} // \oplus \text{denotes the XOR operation} \\ \alpha_i &= \bar{\alpha}_{i-1} \beta_i \\ c_i &= (1 - 2\bar{c}_{i+1})\alpha_i \} \end{aligned}$$

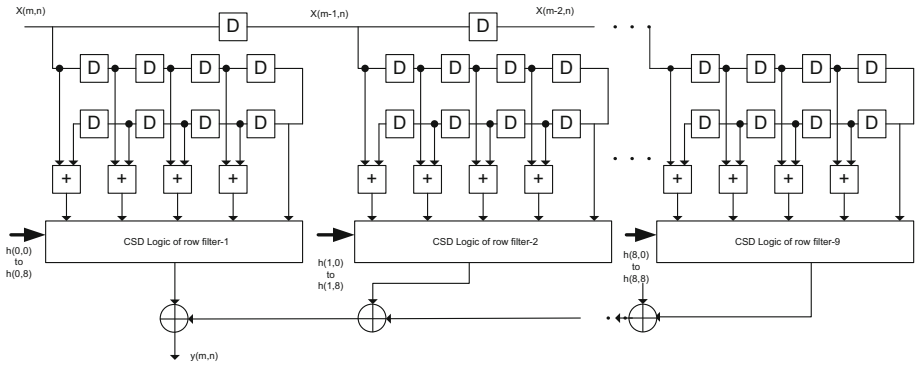


Fig. 6 The architecture of 2D FIR filter in the fully direct form using CSD coefficients

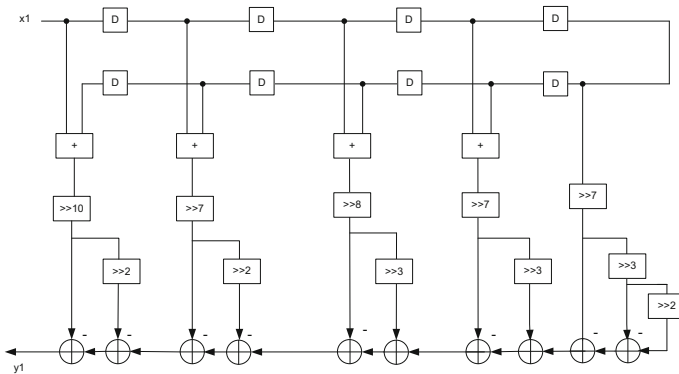


Fig. 7 CSD logic of row filter-1 for fully direct type 2D FIR filter

The constant multiplication is generally carried out by adding or subtracting the number of partial products corresponding to the non-zero elements of the constant multiplier. The CSD representation of the coefficient consists of least number of non-zero bits and hence the number of subtractors and adders to compute the multiplication of input and coefficients are decreased. A 12-bit CSD encoding process has been used for the 2D FIR filter with circular symmetry and CSD coefficients are implemented in fully direct type 2D FIR architecture as shown in Fig. 6. Each row of the two-dimensional coefficients matrix is considered as a sub-filter named as row filter. Each row filter is designed and implemented using CSD format and all the row filters are integrated as direct-form 2D FIR filter structure. The internal hardware structure using CSD logic for row filter-1 is shown in Fig. 7 and the corresponding equation is given in Eq. 14. The CSD format is used to represent the filter coefficients to avoid the multipliers and to reduce the number of adder/subtractors in the filter. Here, the multiplier is completely replaced with add and shift operations only. In this architecture, the row filter structure is implemented using adders, subtractors, and right shifters. According to CSD format, the shifting and adding operations are performed in the applied input signal.

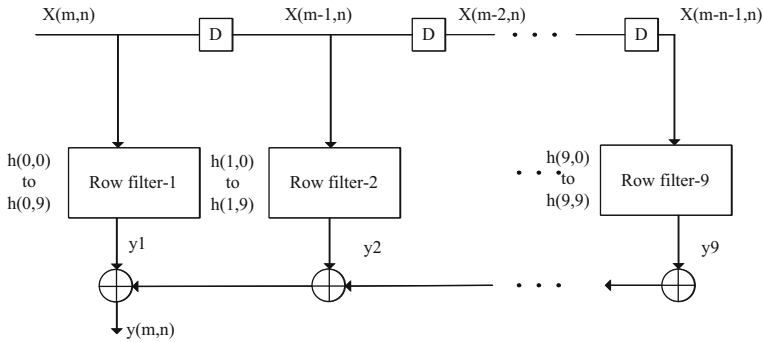


Fig. 8 The architecture of 2D FIR filter in hybrid-II form using CSD coefficients

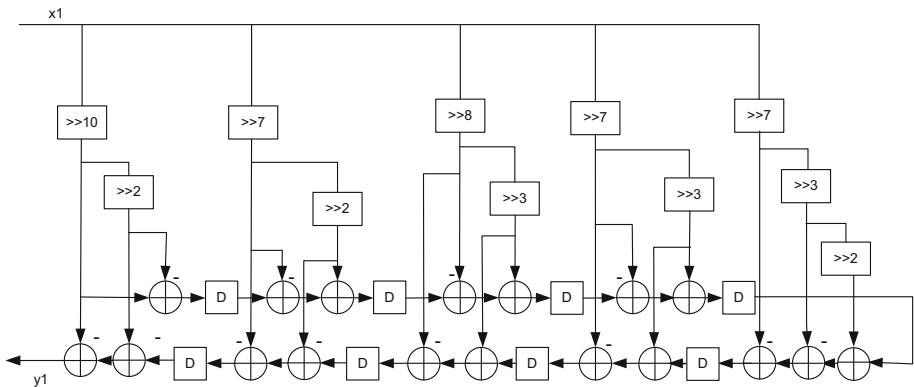


Fig. 9 CSD logic for row filter-1 for hybrid-II type filter

The remaining design equations for 9×9 2D FIR filter with symmetrical row filters are given by Eqs. 15–18.

$$\begin{aligned}
 y1 = y9 = & -[x + (x - 8)] \ggg 10 - [x + (x - 8)] \ggg 12 - [(x - 1) + (x - 7)] \ggg 7 \\
 & - [(x - 1) + (x - 7)] \ggg 9 - [(x - 2) + (x - 6)] \ggg 8 + [(x - 2) + (x - 6)] \ggg 11 \\
 & - [(x - 3) + (x - 5)] \ggg 7 + [(x - 3) + (x - 5)] \\
 & \ggg 10 - (x - 4) \ggg 7 - (x - 4) \ggg 10 + (x - 4) \ggg 12
 \end{aligned} \tag{14}$$

The hybrid-II filter is a combination of direct form and transposed form, which is shown in Fig. 8. Here the registers are placed on the input signal side for overall filter and, inside the row filter the registers are placed on the output side as transposed form. The Fig. 9 represents the internal structure of CSD logic for row filter-1 used in hybrid-II filter architecture.

The output of the row filter-2 and row filter-8 is same and given by Eq. 15.

$$\begin{aligned}
 y2 = y8 = & -[x + (x - 8)] \ggg 7 - [x + (x - 8)] \ggg 9 - [(x - 1) + (x - 7)] \ggg 8 \\
 & - [(x - 2) + (x - 6)] \ggg 10 - [(x - 2) + (x - 6)] \ggg 12 + [(x - 3) + (x - 5)] \ggg 10 \\
 & - [(x - 3) + (x - 5)] \ggg 12 + (x - 4) \ggg 7 - (x - 4) \ggg 9
 \end{aligned} \tag{15}$$

The output of the row filter-3 and row filter-7 are equal and given by Eq. 16.

$$\begin{aligned} y_3 = y_7 = & -[x + (x - 8)] \gg 8 + [x + (x - 8)] \gg 11 - [(x - 1) + (x - 7)] \gg 8 \\ & - [(x - 1) + (x - 7)] \gg 10 - [(x - 1) + (x - 7)] \gg 12 + [(x - 2) + (x - 6)] \gg 6 \\ & - [(x - 2) + (x - 6)] \gg 8 - [(x - 2) + (x - 6)] \gg 10 - [(x - 2) + (x - 6)] \gg 12 \\ & + [(x - 3) + (x - 5)] \gg 5 + [(x - 3) + (x - 5)] \gg 9 - [(x - 3) + (x - 5)] \gg 11 \\ & + (x - 4) \gg 5 + (x - 4) \gg 7 + (x - 4) \gg 9 + (x - 4) \gg 11 \end{aligned} \quad (16)$$

The output of the row filter-4 and row filter-6 are equal and given by Eq. 17.

$$\begin{aligned} y_4 = y_6 = & -[x + (x - 8)] \gg 7 + [x + (x - 8)] \gg 10 + [(x - 1) + (x - 7)] \gg 7 \\ & + [(x - 1) + (x - 7)] \gg 9 - [(x - 1) + (x - 7)] \gg 12 + [(x - 2) + (x - 6)] \gg 5 \\ & + [(x - 2) + (x - 6)] \gg 9 - [(x - 2) + (x - 6)] \gg 11 + [(x - 3) + (x - 5)] \gg 4 \\ & + [(x - 3) + (x - 5)] \gg 6 + [(x - 3) + (x - 5)] \gg 8 + [(x - 3) + (x - 5)] \gg 10 \\ & + [(x - 3) + (x - 5)] \gg 12 + (x - 4) \gg 3 - (x - 4) \gg 5 + (x - 4) \gg 9 \end{aligned} \quad (17)$$

The output of the row filter-5 is given by Eq. 18.

$$\begin{aligned} y_5 = & -[x + (x - 8)] \gg 7 - [x + (x - 8)] \gg 10 + [x + (x - 8)] \gg 12 \\ & + [(x - 1) + (x - 7)] \gg 7 + [(x - 1) + (x - 7)] \gg 9 + [(x - 2) + (x - 6)] \gg 5 \\ & + [(x - 2) + (x - 6)] \gg 7 + [(x - 2) + (x - 6)] \gg 9 + [(x - 2) + (x - 6)] \gg 11 \\ & + [(x - 3) + (x - 5)] \gg 3 - [(x - 3) + (x - 5)] \gg 5 + [(x - 3) + (x - 5)] \gg 9 \\ & + (x - 4) \gg 3 - (x - 4) \gg 7 + (x - 4) \gg 12 \end{aligned} \quad (18)$$

We can observe that, the row filter equations are symmetrical and the multiplication operation is carried out using only adders/subtractors and right shifts operations. Table 2 shows the row filter-1 coefficients in decimal and CSD number with total number of adders 14 and 18 for fully direct form and hybrid-II form filters respectively.

Table 3 presents the number of adders/subtractors required for the implementation of each row filter and the total number of the adders obtained as 151 in fully direct form filter. 66 adders are ignored by symmetric structure then adders are reduced to 85. Table 4 shows the total of 206 adders is required for the entire filter implementation using CSD in hybrid-II form architecture. Here, due to symmetrical, the 206 is reduced to 116, whereas 90 adders are due to the symmetrical operation. The 2D fully direct form filter has 26% of adder saving than the hybrid-II type filter.

3.2 The 2D FIR filter architecture using CSD with common subexpression elimination

The number of adders/subtractors required for each row filter implementation can be further reduced by common subexpression elimination (CSE) technique. The repeated sub-patterns of non-zero bits in one filter coefficient in CSD format can be used for the process of another coefficient. This process will decrease the number of adders required for filter operation (Hsiao et al. 2013). The CSE techniques are two types, Horizontal CSE (HCSE) and Vertical CSE (VCSE).

3.2.1 Horizontal CSE

The common subexpressions are pre-computed and used for the required coefficients. Here, the common non-zero element patterns are identified in the horizontal direction for the entire

Table 2 Comparison of adder complexity for Row filter-1

Row filter-1 coefficients	Decimal value of coefficients	CSD representation	SPT terms	Word length	(Fully direct) No. of adders	(Hybrid-II) no. of adders
$h(0, 0)$	− 0.0012	0.000000000 − 10 − 1	5	12	14	18
$h(0, 1)$	− 0.0098	0.000000 − 10 − 1000				
$h(0, 2)$	− 0.0034	0.0000000 − 100 + 10				
$h(0, 3)$	− 0.0068	0.000000 − 100 + 100				
$h(0, 4)$	− 0.0085	0.000000 − 100 − 10 + 1				
$h(0, 5)$	− 0.0068	0.000000 − 100 + 100				
$h(0, 6)$	− 0.0034	0.0000000 − 100 + 10				
$h(0, 7)$	− 0.0098	0.000000 − 10 − 1000				
$h(0, 8)$	− 0.0012	0.000000000 − 10 − 1				

Table 3 The number of adder/subtractors for all row filters in 2D FIR filter in the fully direct form

Row filter number	No. of adders/subtractors
Row filter-1	14
Row filter-2	12
Row filter-3	20
Row filter-4	20
Row filter-5	19
Row filter-6	20
Row filter-7	20
Row filter-8	12
Row filter-9	14
Total no. of adders	151

row filter as shown in Table 5. The HCSE patterns of row filter-1 and the corresponding internal structure of row filter-1 are shown in the Fig. 10. The Eqs. 19–23 represents the all row filters outputs in 2D FIR filter using HCSE in CSD format. The $h(0, 0)$ coefficient consists of pattern of binary bits (− 1, 0, − 1) and same pattern is repeated in coefficients $h(0, 1)$, $h(0, 7)$ and $h(0, 8)$ which are highlighted in Table 4. Similarly, the pattern (− 1, 0, 0, + 1) bits in $h(0, 2)$ also repeated in $h(0, 3)$, $h(0, 5)$ and $h(0, 6)$ coefficients.

The row filter-1 and row filter-9 are symmetrical concerning hardware as shown in the Eq. 19.

The common subexpressions from Table 5 are assumed as x_2 and x_3 are given by,

Table 4 Number of adder/subtractors for all row filters in 2D FIR filter in hybrid-II form

Row filter number	No. of adders/subtractors
Row filter-1	18
Row filter-2	17
Row filter-3	27
Row filter-4	28
Row filter-5	26
Row filter-6	28
Row filter-7	27
Row filter-8	17
Row filter-9	18
Total no. of adders	206

Table 5 Row filter-1 with 12-bit CSD coefficients using HCSE

Coefficients	CSD representation	1	2	3	4	5	6	7	8	9	10	11	12
$h(0, 0)$	0.000000000 - 10 - 1	0	0	0	0	0	0	0	0	0	-1	0	-1
$h(0, 1)$	0.0000000 - 10 - 1000	0	0	0	0	0	0	-1	0	-1	0	0	0
$h(0, 2)$	0.0000000 - 100 + 10	0	0	0	0	0	0	0	-1	0	0	+1	0
$h(0, 3)$	0.0000000 - 100 + 100	0	0	0	0	0	0	-1	0	0	+1	0	0
$h(0, 4)$	0.0000000 - 100 - 10 + 1	0	0	0	0	0	0	-1	0	0	-1	0	+1
$h(0, 5)$	0.0000000 - 100 + 100	0	0	0	0	0	0	-1	0	0	+1	0	0
$h(0, 6)$	0.0000000 - 100 + 10	0	0	0	0	0	0	0	-1	0	0	+1	0
$h(0, 7)$	0.0000000 - 10 - 1000	0	0	0	0	0	0	-1	0	-1	0	0	0
$h(0, 8)$	0.000000000 - 10 - 1	0	0	0	0	0	0	0	0	0	-1	0	-1

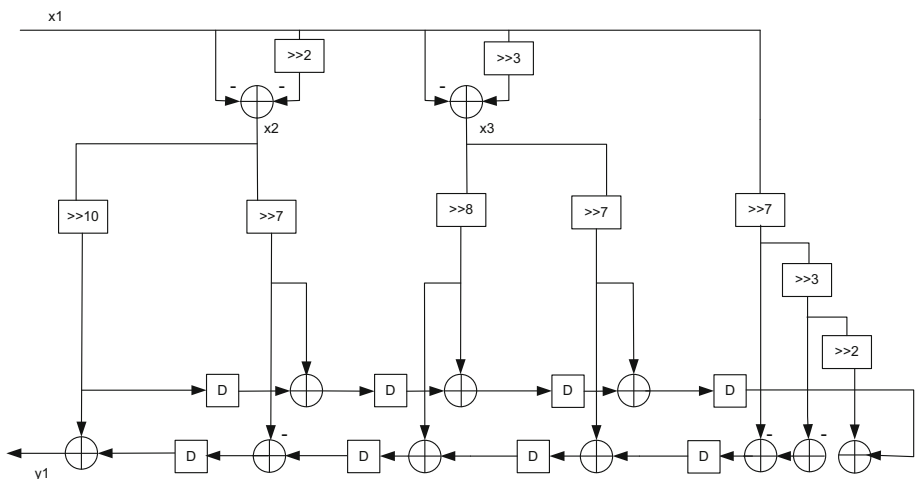


Fig. 10 Hardware structure of row filter-1 using HCSE

Assume $x_2 = -x_1 - x_1 \gg 2$ and $x_3 = -x_1 + x_1 \gg 3$.

$$y_1 = y_9 = [x_2 + (x_2 - 8)] \gg 10 + [(x_2 - 1) + (x_2 - 7)] \gg 7 + [(x_3 - 2) + (x_3 - 6)] \gg 8 \\ + [(x_3 - 3) + (x_3 - 5)] \gg 7 - (x_1 - 4) \gg 7 - (x_1 - 4) \gg 10 + (x_1 - 4) \gg 12 \quad (19)$$

Assume $x_2 = -x_1 - x_1 \gg 2$ and $x_3 = x_1 - x_1 \gg 2$, then the output of the row filter-2 is same as row filter-8 is given by,

$$y_2 = y_8 = [x_2 + (x_2 - 8)] \gg 7 - [(x_1 - 1) + (x_1 - 7)] \gg 8 - [(x_1 - 2) + (x_1 - 6)] \gg 8 \\ + [(x_2 - 2) + (x_2 - 6)] \gg 10 + [(x_3 - 3) + (x_3 - 5)] \gg 10 - (x_3 - 4) \gg 7 \quad (20)$$

Assume $x_2 = -x_1 - x_1 \gg 2$ and $x_3 = x_1 + x_1 \gg 2$, then the row filter-3 and row filter-7 output is same and is given by,

$$y_3 = y_7 = -[x_1 + (x_1 - 8)] \gg 8 + [x_1 + (x_1 - 8)] \gg 11 + [x_2 + (x_2 - 7)] \gg 8 \\ + [(x_1 - 2) + (x_1 - 6)] \gg 6 + [(x_2 - 2) + (x_2 - 6)] \gg 8 \\ + [(x_1 - 3) + (x_1 - 5)] \gg 5 + [(x_1 - 3) + (x_1 - 5)] \gg 9 \\ - [(x_1 - 3) + (x_1 - 5)] \gg 11 + (x_3 - 4) \gg 5 + (x_3 - 4) \gg 9 \quad (21)$$

Assume $x_2 = -x_1 + x_1 \gg 3$ and $x_3 = x_1 - x_1 \gg 2$ then the row filter-4 output is same as row filter-6 is same is given by,

$$y_4 = y_6 = [x_2 + (x_2 - 8)] \gg 7 + [(x_1 - 1) + (x_1 - 7)] \gg 7 - [(x_2 - 1) + (x_2 - 7)] \gg 9 \\ + [(x_1 - 2) + (x_1 - 6)] \gg 5 + [(x_3 - 2) + (x_3 - 6)] \gg 9 + [(x_1 - 3) + (x_1 - 5)] \gg 4 \\ + [(x_3 - 3) + (x_3 - 5)] \gg 6 + [(x_1 - 3) + (x_1 - 5)] \gg 10 + [(x_1 - 3) + (x_1 - 5)] \gg 12 \\ + (x_3 - 4) \gg 3 + (x_1 - 4) \gg 9 \quad (22)$$

Assume $x_2 = -x_1 + x_1 \gg 2$ and $x_3 = x_1 + x_1 \gg 2$ then the row filter-5 output is,

$$y_5 = -[x + (x - 8)] \gg 7 + [(x_2 + (x_2 - 8)] \gg 10 - [(x_2 - 1) + (x_2 - 7)] \gg 7 \\ + [(x_3 - 2) + (x_3 - 6)] \gg 5 + [(x_3 - 2) + (x_3 - 6)] \gg 9 - [(x_2 - 3) + (x_2 - 5)] \gg 3 \\ + [(x - 3) + (x - 5)] \gg 9 + (x - 4) \gg 3 - (x - 4) \gg 7 + (x - 4) \gg 12 \quad (23)$$

The coefficient implementation of row filter-1 in CSD required 17 adder/subtractors, whereas in CSD with HCSE required only 12 adders. HCSE has been reduced by 5 adders for single row filter. The total number of adders for the 2D FIR filter using HCSE in CSD format is 146 as shown in Table 6. Due to symmetry, 64 are adders are subtracted from the total number of adders, then the resultant adder number is 82 only. The original CSD architecture required a total of 116 adders/subtractors for hybrid-II architecture and whereas CSD with HCSE required only 82 adder/subtractors. Hence, 29% of adder/subtractors are saved by HCSE in CSD architecture.

3.2.2 Vertical CSE

VCSE also can be used for the further reduction of adders in coefficient implementation instead of horizontal CSE. It may not continue the linear phase filter symmetry. It is suitable for the hybrid-II filter structure. Sometimes, particularly in a few row filters, the number of adders may be maintained as same as HCSE or more than.

The row filter-4 is considered to understand the technique of VCSE in CSD as shown in Table 7. The vertical common patterns are highlighted and used to derive the design Eq. 24. The row filter-4 consumes 19 adders and whereas in CSD with HCSE, 21 adders are required

Table 6 Number of adders/subtractors for all row filters in 2D FIR architecture using CSD-HCSE

Row filter number	HCSE No. of adders/subtractors
Row filter-1	12
Row filter-2	12
Row filter-3	19
Row filter-4	21
Row filter-5	18
Row filter-6	21
Row filter-7	19
Row filter-8	12
Row filter-9	12
Total no. of adders	146

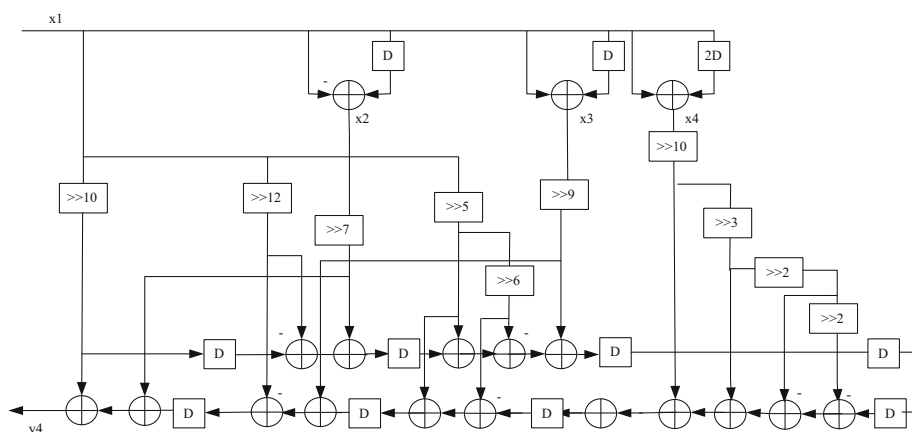


Fig. 11 Logic structure for row filter-4 using CSD with VCSE

to implementation of row filter-4. The hardware structure of row filter-4 using CSD-VCSE is shown in Fig. 11.

From the Table 7, the common sub expressions are assumed as follows:

$$x2 = -x1 + (x1 - 1),$$

$$x3 = x1 + (x1 - 1),$$

$$x4 = x1 + (x1 - 2)$$

Then the output equation of row filter-4 is,

$$\begin{aligned}
 y4 = & [x1 + (x1 - 8)] \gg 10 - [(x1 - 1) + (x1 - 7)] \gg 12 + [(x1 - 2) + (x1 - 6)] \gg 5 \\
 & - [(x1 - 2) + (x1 - 6)] \gg 11 + [x2 - (x2 - 7)] \gg 7 + [(x3 - 1) + (x3 - 6)] \gg 9 \\
 & + (x4 - 3) \gg 4 + (x4 - 3) \gg 8 - (x4 - 3) \gg 10 - (x4 - 3) \gg 12 \quad (24)
 \end{aligned}$$

All the 9 row filters implemented using VCSE in CSD technique. Table 8 represents the total number of adders for the implementation of 2D FIR filter using CSD with VCSE is 153 and symmetry reduce the 67 adders. Therefore the net number of adders is 86. Where

Table 7 Row filter-4 with 12-bit CSD coefficients using VCSE

Coefficients	CSD representation	1	2	3	4	5	6	7	8	9	10	11	12
$h(0, 0)$	0.000000 - 100 + 100	0	0	0	0	0	0	-1	0	0	+1	0	0
$h(0, 1)$	0.000000 + 10 + 100 - 1	0	0	0	0	0	0	+1	0	+1	0	0	-1
$h(0, 2)$	0.0000 + 1000 + 10 - 10	0	0	0	0	+1	0	0	0	+1	0	-1	0
$h(0, 3)$	0.000 + 10 + 10 - 10 - 10 - 1	0	0	0	+1	0	+1	0	-1	0	-1	0	-1
$h(0, 4)$	0.00 + 10 - 1000 + 1000	0	0	+1	0	-1	0	0	0	+1	0	0	0
$h(0, 5)$	0.000 + 10 + 10 - 10 - 10 - 1	0	0	0	+1	0	+1	0	-1	0	-1	0	-1
$h(0, 6)$	0.0000 + 1000 + 10 - 10	0	0	0	0	+1	0	0	0	+1	0	-1	0
$h(0, 7)$	0.000000 + 10 + 100 - 1	0	0	0	0	0	0	+1	0	+1	0	0	-1
$h(0, 8)$	0.000000 - 100 + 100	0	0	0	0	0	0	-1	0	0	+1	0	0

Table 8 Number of adders/subtractors for all row filters in 2D FIR architecture in VCSE

Row filter number	VCSE no. of adders/subtractors
Row filter-1	16
Row filter-2	13
Row filter-3	20
Row filter-4	18
Row filter-5	19
Row filter-6	18
Row filter-7	20
Row filter-8	13
Rowfilter-9	16
Total no. of adders	153

HCSE in CSD space requires 82 only and CSD requires 116 respectively. VCSE reduce the 25% adders/subtractors comparatively CSD architecture. Some significant saving in adders achieved using HCSE as well as VCSE in CSD format of filter architectures.

4 Experimental results

The circular symmetric 2D FIR filter is designed using kth-McClellan transform with T2 and P2 transformations in MATLAB. The fully direct form and hybrid-II form architectures are implemented using generic multipliers for the continuous coefficients with CAVM and MCM techniques respectively in Hardware Description Language (HDL). The continuous coefficients are converted into CSD number format to implement multiplierless architectures. The optimization is introduced in CSD using CSE algorithms to reduce the complexity in hardware.

The reconfigurable designs are implemented using Xilinx ISE tool with Verilog HDL and the custom designs are implemented in TSMC 45 nm CMOS generic library using RTL encounter from CADENCE tools. The fully direct and hybrid-II filter architectures are proposed to meet the desired specifications in multiplierless multiplication technique.

For reconfigurable applications, Vertex-6 FPGA is considered as target device and both the architectures of 2D FIR filters are implemented. For comparison purpose, architectures are also implemented for continuous coefficients using generic multipliers with MCM and CAVM schemes. The hardware complexity comparison concerning target device is FPGA is presented in Table 9. The continuous coefficients design requires DSP48 slices for the computation of the multiplier. For the multiplierless design, no DSP48 slices were used and these resources may be used for the other computations in the application. The multiplierless design is applied for the two architectures fully direct form and hybrid-II form. The fully direct filter requires less number of slices and (Look-Up-Table) LUTs comparatively hybrid-II filter. Next, the CSD and CSD with CSE decreases the number of adders/subtractors and registers also. The power dissipation and delay of both architectures is directly decreased. The different input bit widths are considered for the filter with multipliers and multiplierless 2D filter architectures. The input bit width is reduced from 16 to 8 for the continuous coefficient 2D filter, then the output bit width also reduced. This can prune the complexity of the filter in terms of area, power, and delay. This is because the input bit width affects the size of the adder/subtractors, multipliers, and registers.

Table 9 Comparison of complexity and parameters of 2D FIR filter in different architectures with target device of FPGA

Type of coefficients	Type of architecture	8-bit filter input			16-bit filter input				
		LUTs	FFs	Slice Registers	DSP48Es	LUTs	FFs	Slice Registers	DSP48Es
Continuous coefficients	Fully Direct	2048	387	416	90	2549	481	562	90
	Hybrid-II	2056	598	578	90	2688	715	688	90
CSD	Fully Direct	624	325	398	0	2549	987	1002	0
	Hybrid-II	651	373	416	0	2691	1010	1097	0
CSD with HCSE	Fully Direct	560	321	335	0	2002	866	897	0
	Hybrid-II	563	365	379	0	2013	895	989	0
CSD with VCSE	Fully Direct	588	332	397	0	2022	864	872	0
	Hybrid-II	681	381	412	0	2076	912	897	0

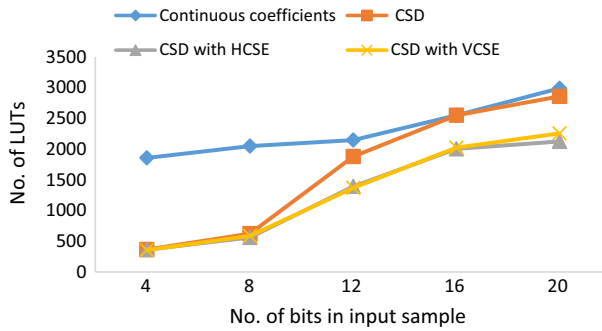


Fig. 12 Comparison of number of LUTs required for different coefficient representations

Figure 12 describes the comparison of number of LUTs required for the implementation of fully direct architecture for different input widths, such as 4, 8, 12, 16 and 20 bits. The number of LUTs required for CSD with HCSE almost same as CSD with VCSE. But less number of LUTs when compared with conventional CSD and continuous coefficients architectures.

4.1 Custom design results

The (Application Specific Integrated Circuit) ASIC or Custom design of the multiplierless filters both fully direct and hybrid-II form 2D filter architectures leads to savings in the area and power to the continuous coefficient 2D FIR filter. The complexity of the fully direct form filter is less than the hybrid-II filter. The reason for this, the size of the registers in the hybrid form is increasing after multiplication process, whereas in the direct form filter, the bit width and the corresponding number of registers are same at the input and output sides. For the implementation of each row filter of direct form required less number of adders than the transposed form. The delay of the continuous coefficient implementation with generic multipliers is less than the shift-add designs due to more operations are required and connected in series in the multiplierless design. To reduce this delay to some extent Carry Save adders are used for filter structural adders.

Table 10 presents the four different coefficient types and two forms of architectures for 2D filters comparison in terms of area, power and delay with different input bit widths. The both proposed 2D circular symmetric FIR filters in CSD with HCSE has given best results for desired filter specifications. The multiplierless design of fully direct form 2D FIR filter with CSD-HCSE given 90% power reduction when compared with continuous filter coefficients with multipliers. The delay is reduced by 8% using CSD, CSD with HCSE and CSD with VCSE architectures when compared to continuous coefficient filter architecture. More than 80% area saving is obtained by multiplierless designs using CSD, CSD-HCSE, and CSD-VCSE than the conventional continuous coefficient filter.

The Figs. 13 and 14 represents the line graph comparison of power consumption and area of CSD filter, conventional continuous coefficients, CSD-HCSE and CSD-VCSE with respect to different number of bits per input sample respectively on ASIC platform. From the observations, CSD architectures power consumption and area are very much less as compared with continuous coefficients structure.

In Table 11, the proposed circular symmetric 2D FIR filter architectures are compared with the existing 2D FIR filter architectures with the nearest length of the filter in terms of VLSI

Table 10 Comparison of complexity and parameters of 2D FIR filter in different architectures in Custom design or ASIC platform

Type of coefficients	Type of architecture	8-bit filter input			16-bit filter input				
		Cells	Area (μm^2)	Power (mW)	Delay (ns)	Cells	Area (μm^2)	Power (mW)	Delay (ns)
Continuous Coefficients	Fully Direct	7899	89,010	34.8	905	16,980	18,9543	51.002	1800
	Hybrid-II	7299	90,134	35.60	905	16,188	188,319	50.84	1809
CSD	Fully Direct	872	3109	2.1803	628	5347	16,239	5.1358	1656
	Hybrid-II	951	3305	2.3041	611	5920	18,049	5.3304	1523
CSD with HCSE	Fully Direct	824	2429	2.309	623	5009	16,105	5.1204	1640
	Hybrid-II	876	2921	2.3234	598	5437	17,809	5.2432	1500
CSD with VCSE	Fully Direct	799	2571	2.7898	635	5231	16,111	5.7538	1650
	Hybrid-II	822	2989	2.9001	605	5333	17,635	5.8390	1512

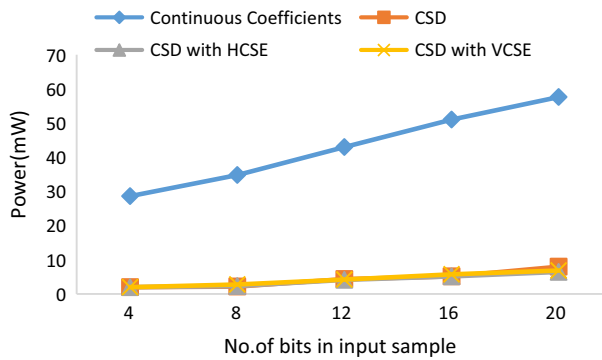


Fig. 13 Comparison of power consumption of different coefficient representations

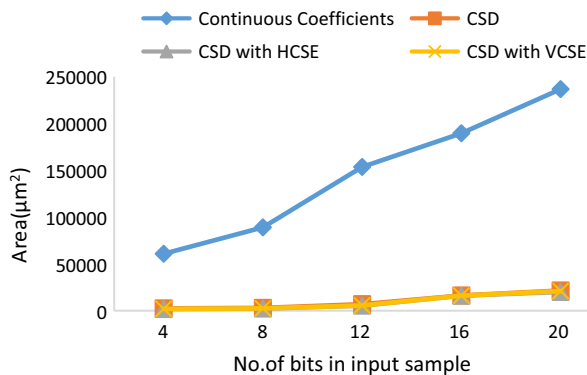


Fig. 14 Comparison of area occupancy of different coefficient representations of 2D FIR filters

Table 11 Comparison of synthesis results of proposed 2D FIR filters with state-of-art works

Architectures	N (Order of the filter)	Area (μm^2)	Power (mW)	Delay (ns)	ADP (μm^2 ms)
Kumar et al. (2018)	8	651,615	20.1069	6.53	0.531880
Mohanty et al. (2013)	8	1,720,962	50.0106	11.79	25.375558
Khoo et al. (2010)	8	356,293	9.3807	13.1	4.636084
Chen et al. (2010)	9	22,359	9.356	19.23	0.429963
Proposed 1 (Fully Direct)	9	17,199	5.1204	16.52	0.284127
Proposed 2 (Hybrid-II)	9	17,809	5.2432	16.56	0.294917

N is order of the filter, ADP= Area Delay Product

parameters. Total 75% and 85% of power saving is obtained by proposed filter architectures when compared with 2D FIR filter architecture proposed in Kumar et al. (2018) and Mohanty et al. (2013) respectively. More than 45% of power saving is obtained using the proposed-1 2D FIR filter than Khoo et al. (2010) and Chen et al. (2010) for fixed desired coefficients. This power saving achieved by replacing the power hunger multiplier blocks with shift-add technique using CSD representation of coefficients in 2D filters.

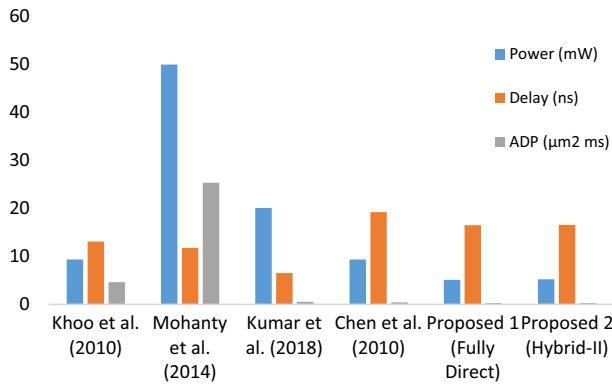


Fig. 15 Comparison of proposed architectures with existing works in terms of delay, power and ADP for N=8

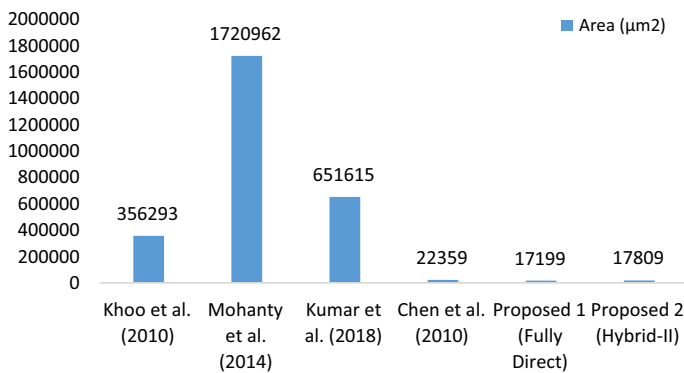


Fig. 16 Comparison of proposed architecture area with existing works for N=8

Drastic reduction in area is achieved by proposed filters when comparing with 2D FIR filter architectures of Kumar et al. (2018), Mohanty et al. (2013) and Khoo et al. (2010). The 47% and 39% of ADP is reduced when compared with ADP results of Kumar et al. (2018) and Chen et al. (2010) respectively.

The ADP values of Mohanty et al. (2013) and Khoo et al. (2010) are huge when compared to proposed architectures.

The two proposed 8th order filter architectures are compared with four different state-of-art works in terms of power, delay and Area Delay Product (ADP) values using bar graphs in Fig. 15. It may be noted that the power and ADP of proposed architectures are very much less than when compared with existing filter architectures. Figure 16 depicts the area comparison of existing and proposed filter structures.

The proposed CSD-HCSE and CSD-VCSE filter architectures are also implemented for N=16 and the detailed comparison results of filter architectures are presented in Table 12. The Fig. 17 depicts the comparison of power, delay and ADP of proposed filters with existing filters for N=16. From Table 12, it can be observed that Fully direct architecture with N=16 are more efficient in terms of power and ADP by 89% and 52% respectively, when compared with DA based block structure Kumar et al. (2018). The huge power reduction and ADP reduction is achieved in proposed methods when compared with Mohanty et al. (2013), Meher and Park (2011) and Guo and DeBrunner (2011). This is due to the optimization using CSE in

Table 12 Comparison of synthesis results of existing and proposed 2D FIR filters for N=16

Architectures	N (Order of the filter)	Area (μm^2)	Power (mW)	Delay (ns)	ADP (μm^2 ms)
Kumar et al. (2018)	16	4,936,081	102.3432	6.91	2.131770
Mohanty et al. (2013)	16	12,661,783	181.165	13.33	10.550431
Meher and Park (2011)	16	59,244	12.08	24.38	1.444368
Guo and DeBrunner (2011)	16	40,187	21.16	12.92	0.519216
Khoo et al. (2010)	16	1,154,123	26.9941	14.65	16.907903
Proposed 1 (Fully Direct)	16	54,321	10.2354	18.98	1.031012
Proposed 2 (Hybrid-II)	16	56,243	10.5689	19.06	1.071991

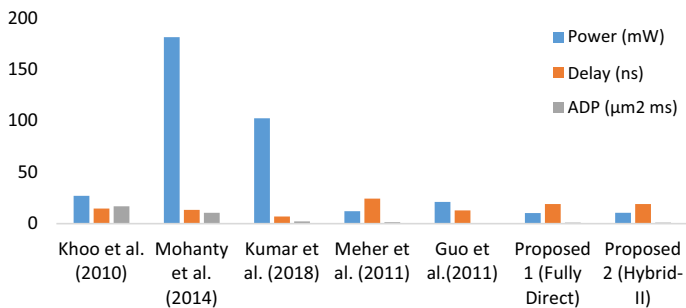


Fig. 17 Comparison of ADP, delay and power for filter length N=16

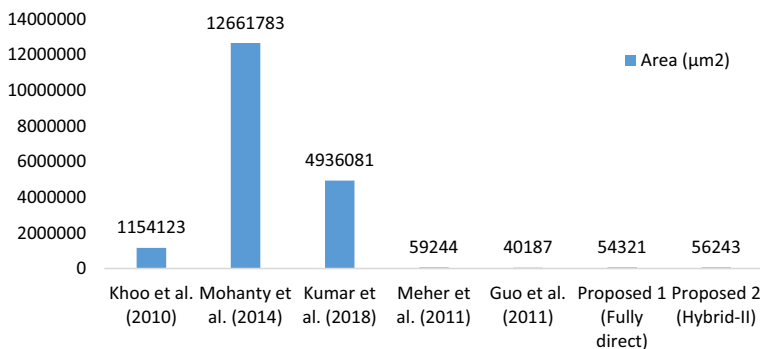


Fig. 18 Comparison of area for proposed architecture with existing works (N=16)

CSD technique. It may be observed that from Fig. 18 the significant area saving is obtained than other state-of-art architectures.

Figure 19 shows the power comparison of proposed architectures with architectures available in the literature for the filter order of N = 8 and N = 16. It is observed that the power is proportionally increasing with respect to order of the filter in existing architectures. But in the proposed filter architectures less power variations are observed with increase in order of

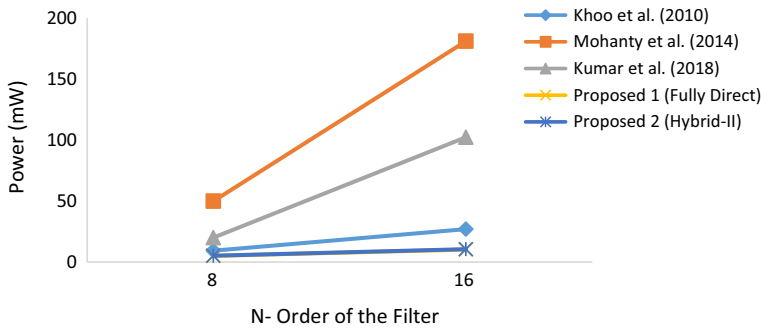


Fig. 19 Comparison of power consumption for various filter architectures for $N = 8$ and $N = 6$

the filter. The propose filter 1(Fully Direct form) and proposed 2 filter (Hybrid-II) consumes almost same power.

5 Conclusion

In this paper, the prominent architectures are designed and implemented using multiplierless constant multiplications and described in fully direct form and hybrid-II form architectures of 2D FIR filter. The filter coefficients are generated for the desired specifications to achieve circular symmetry using kth-Park's McClellan transformation, T2 and P2 transformation techniques. The continuous coefficients are optimized using a CSD number format to reduce the number of adder/subtractors. Further, the optimization of filter coefficients is carried out by CSE algorithms. Horizontal and Vertical CSE techniques have been used to reduce the hardware complexity of CSD. A low complexity circular symmetric 2D FIR filter is proposed, which has been designed in MATLAB and implemented in TSMC 45 nm CMOS technology using the generic library. High-level synthesis is carried out for the continuous filter with generic multipliers and multiplierless designs with CSD, CSD-HCSE, and CSD-VCSE.

For area-efficient applications, the fully direct form is the best architecture to implement symmetric 2D FIR filters in custom designs and FPGA designs. For high-speed applications, the hybrid-II form is the best architecture on Custom designs and reconfigurable design platform. For low power applications, the algorithms to optimize hardware complexity is better to implement the filter in a fully direct form in ASIC platform. The experimental results indicate that high-level synthesis algorithms have a significant impact on the multiplierless design of 2D FIR filters in terms of complexity, power, and performance using both architectures. The complexity comparison and analysis is carried out for two different filter orders such as $N=8$ and $N=16$. The limitation of the filter design is fixed coefficients are considered for certain specifications only.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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